$$
\begin{aligned}
C_{ll}{}^{i1\_i2\_i3\_i4\_} \to \frac{1}{16\pi^2} \Bigg(& \frac{1}{288}\frac{1}{m_\Phi^2} s_\gamma^2 \left(\overline{y_e}^{i4p}\left(y_e^{i3p}\left(-7g_1^2+g_2^2\right)\delta_{i1i2}-2g_2^2 y_e^{i1p}\delta_{i2i3}\right)+\right.\\
& \left.\overline{y_e}^{i2p}\left(-2y_e^{i3p}\left(18c_\gamma^2\overline{y_e}^{i4r}y_e^{i1r}+g_2^2\delta_{i1i4}\right)+y_e^{i1p}\left(-7g_1^2+g_2^2\right)\delta_{i3i4}\right)\right)+\\
& \frac{1}{18}g_2^4\left(2\delta_{i1i4}\delta_{i2i3}-\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{3,0}[m_2]+\frac{1}{12}g_2^4\left(2\delta_{i1i4}\delta_{i2i3}-\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{4,-1}[m_2]+\\
& \frac{4}{45}g_2^4\left(-2\delta_{i1i4}\delta_{i2i3}+\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{5,-2}[m_2]+\\
& \frac{1}{54}\sum_p g_1^4\,\text{LF}_{3,0}\left[m_{\tilde{d}}^p\right]\delta_{i1i2}\delta_{i3i4}-\frac{5}{144}\sum_p g_1^4\,\text{LF}_{4,-1}\left[m_{\tilde{d}}^p\right]\delta_{i1i2}\delta_{i3i4}+\\
& \frac{2}{135}\sum_p g_1^4\,\text{LF}_{5,-2}\left[m_{\tilde{d}}^p\right]\delta_{i1i2}\delta_{i3i4}+\frac{1}{18}\sum_p g_1^4\,\text{LF}_{3,0}\left[m_{\tilde{e}}^p\right]\delta_{i1i2}\delta_{i3i4}-\\
& \frac{5}{48}\sum_p g_1^4\,\text{LF}_{4,-1}\left[m_{\tilde{e}}^p\right]\delta_{i1i2}\delta_{i3i4}+\frac{2}{45}\sum_p g_1^4\,\text{LF}_{5,-2}\left[m_{\tilde{e}}^p\right]\delta_{i1i2}\delta_{i3i4}+\\
& \frac{1}{36}\sum_p\left(2g_2^4\delta_{i1i4}\delta_{i2i3}+\left(g_1^4-g_2^4\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{3,0}\left[m_{\tilde{l}}^p\right]-\\
& \frac{5}{96}\sum_p\left(2g_2^4\delta_{i1i4}\delta_{i2i3}+\left(g_1^4-g_2^4\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{4,-1}\left[m_{\tilde{l}}^p\right]+\\
& \frac{1}{45}\sum_p\left(2g_2^4\delta_{i1i4}\delta_{i2i3}+\left(g_1^4-g_2^4\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{5,-2}\left[m_{\tilde{l}}^p\right]+\\
& \frac{1}{108}\sum_p\left(18g_2^4\delta_{i1i4}\delta_{i2i3}+\left(g_1^4-9g_2^4\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{3,0}\left[m_{\tilde{q}}^p\right]-\\
& \frac{5}{288}\sum_p\left(18g_2^4\delta_{i1i4}\delta_{i2i3}+\left(g_1^4-9g_2^4\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{4,-1}\left[m_{\tilde{q}}^p\right]+\\
& \frac{1}{135}\sum_p\left(18g_2^4\delta_{i1i4}\delta_{i2i3}+\left(g_1^4-9g_2^4\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{5,-2}\left[m_{\tilde{q}}^p\right]+\\
& \frac{2}{27}\sum_p g_1^4\,\text{LF}_{3,0}\left[m_{\tilde{u}}^p\right]\delta_{i1i2}\delta_{i3i4}-\frac{5}{36}\sum_p g_1^4\,\text{LF}_{4,-1}\left[m_{\tilde{u}}^p\right]\delta_{i1i2}\delta_{i3i4}+\\
& \frac{8}{135}\sum_p g_1^4\,\text{LF}_{5,-2}\left[m_{\tilde{u}}^p\right]\delta_{i1i2}\delta_{i3i4}-\\
& \frac{1}{12}s_\gamma^2\left(g_1^2\overline{y_e}^{i4p}y_e^{i3p}\delta_{i1i2}+\overline{y_e}^{i2p}\left(-3c_\gamma^2\overline{y_e}^{i4r}y_e^{i1r}y_e^{i3p}+g_1^2y_e^{i1p}\delta_{i3i4}\right)\right)\text{LF}_{1,2}[m_\Phi]+\\
& \frac{1}{48}s_\gamma^2\left(\overline{y_e}^{i4p}\left(y_e^{i3p}\left(g_1^2+g_2^2\right)\delta_{i1i2}-2g_2^2y_e^{i1p}\delta_{i2i3}\right)+\right.\\
& \left.\overline{y_e}^{i2p}\left(2y_e^{i3p}\left(3s_\gamma^2\overline{y_e}^{i4r}y_e^{i1r}-g_2^2\delta_{i1i4}\right)+y_e^{i1p}\left(g_1^2+g_2^2\right)\delta_{i3i4}\right)\right)\text{LF}_{2,1}[m_\Phi]+\\
& \frac{1}{144}\left(8g_2^4\delta_{i1i4}\delta_{i2i3}+3s_\gamma^2\overline{y_e}^{i4p}\left(-y_e^{i3p}\left(g_1^2+g_2^2\right)\delta_{i1i2}+2g_2^2y_e^{i1p}\delta_{i2i3}\right)+\right.\\
& \left.4\left(g_1^4-g_2^4\right)\delta_{i1i2}\delta_{i3i4}+3s_\gamma^2\overline{y_e}^{i2p}\left(2g_2^2y_e^{i3p}\delta_{i1i4}-y_e^{i1p}\left(g_1^2+g_2^2\right)\delta_{i3i4}\right)\right)\text{LF}_{3,0}[m_\Phi]+\\
& \frac{1}{96}\left(-10g_2^4\delta_{i1i4}\delta_{i2i3}+5\left(-g_1^4+g_2^4\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{4,-1}[m_\Phi]+\\
& \frac{1}{45}\left(2g_2^4\delta_{i1i4}\delta_{i2i3}+\left(g_1^4-g_2^4\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{5,-2}[m_\Phi]+\\
& \frac{1}{36}\left(2g_2^4\delta_{i1i4}\delta_{i2i3}+\left(g_1^4-g_2^4\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{3,0}[\tilde{\mu}]+\\
& \frac{1}{24}\left(2g_2^4\delta_{i1i4}\delta_{i2i3}+\left(g_1^4-g_2^4\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{4,-1}[\tilde{\mu}]+\\
& \frac{1}{45}\left(-4g_2^4\delta_{i1i4}\delta_{i2i3}+2\left(-g_1^4+g_2^4\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{5,-2}[\tilde{\mu}]+\\
& \frac{1}{192}g_1^2\left(2g_2^2\delta_{i1i4}\delta_{i2i3}+\left(g_1^2-g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{2,1,0}\left[m_1,m_{\tilde{l}}^{i1}\right]+\\
& \frac{1}{192}g_1^2\left(2g_2^2\delta_{i1i4}\delta_{i2i3}+\left(g_1^2-g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{2,2,-1}\left[m_1,m_{\tilde{l}}^{i1}\right]+\\
& \frac{1}{96}g_1^2\left(-2g_2^2\delta_{i1i4}\delta_{i2i3}+\left(-g_1^2+g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{3,1,-1}\left[m_1,m_{\tilde{l}}^{i1}\right]+\\
& \frac{1}{192}g_1^2\left(2g_2^2\delta_{i1i4}\delta_{i2i3}+\left(g_1^2-g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{4,1,-2}\left[m_1,m_{\tilde{l}}^{i1}\right]+\\
& \frac{1}{192}g_1^2\left(2g_2^2\delta_{i1i4}\delta_{i2i3}+\left(g_1^2-g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{2,1,0}\left[m_1,m_{\tilde{l}}^{i2}\right]+\\
& \frac{1}{192}g_1^2\left(2g_2^2\delta_{i1i4}\delta_{i2i3}+\left(g_1^2-g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{2,2,-1}\left[m_1,m_{\tilde{l}}^{i2}\right]+\\
& \frac{1}{96}g_1^2\left(-2g_2^2\delta_{i1i4}\delta_{i2i3}+\left(-g_1^2+g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{3,1,-1}\left[m_1,m_{\tilde{l}}^{i2}\right]+\\
& \frac{1}{192}g_1^2\left(2g_2^2\delta_{i1i4}\delta_{i2i3}+\left(g_1^2-g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{4,1,-2}\left[m_1,m_{\tilde{l}}^{i2}\right]+\\
& \frac{1}{192}g_1^2\left(2g_2^2\delta_{i1i4}\delta_{i2i3}+\left(g_1^2-g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{2,1,0}\left[m_1,m_{\tilde{l}}^{i3}\right]+\\
& \frac{1}{192}g_1^2\left(2g_2^2\delta_{i1i4}\delta_{i2i3}+\left(g_1^2-g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{2,2,-1}\left[m_1,m_{\tilde{l}}^{i3}\right]+\\
& \frac{1}{96}g_1^2\left(-2g_2^2\delta_{i1i4}\delta_{i2i3}+\left(-g_1^2+g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{3,1,-1}\left[m_1,m_{\tilde{l}}^{i3}\right]+\\
& \frac{1}{192}g_1^2\left(2g_2^2\delta_{i1i4}\delta_{i2i3}+\left(g_1^2-g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{4,1,-2}\left[m_1,m_{\tilde{l}}^{i3}\right]+\\
& \frac{1}{192}g_1^2\left(2g_2^2\delta_{i1i4}\delta_{i2i3}+\left(g_1^2-g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{2,1,0}\left[m_1,m_{\tilde{l}}^{i4}\right]+\\
& \frac{1}{192}g_1^2\left(2g_2^2\delta_{i1i4}\delta_{i2i3}+\left(g_1^2-g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{2,2,-1}\left[m_1,m_{\tilde{l}}^{i4}\right]+\\
& \frac{1}{96}g_1^2\left(-2g_2^2\delta_{i1i4}\delta_{i2i3}+\left(-g_1^2+g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{3,1,-1}\left[m_1,m_{\tilde{l}}^{i4}\right]+\\
& \frac{1}{192}g_1^2\left(2g_2^2\delta_{i1i4}\delta_{i2i3}+\left(g_1^2-g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{4,1,-2}\left[m_1,m_{\tilde{l}}^{i4}\right]+\\
& \frac{1}{192}\left(-2g_2^4\delta_{i1i4}\delta_{i2i3}+g_2^2\left(3g_1^2+g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{2,1,0}\left[m_2,m_{\tilde{l}}^{i1}\right]+\\
& \frac{1}{192}\left(-2g_2^4\delta_{i1i4}\delta_{i2i3}+g_2^2\left(3g_1^2+g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{2,2,-1}\left[m_2,m_{\tilde{l}}^{i1}\right]+\\
& \frac{1}{96}\left(-10g_2^4\delta_{i1i4}\delta_{i2i3}+g_2^2\left(-3g_1^2+5g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{3,1,-1}\left[m_2,m_{\tilde{l}}^{i1}\right]+\\
& \frac{1}{64}\left(2g_2^4\delta_{i1i4}\delta_{i2i3}+g_2^2\left(g_1^2-g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{4,1,-2}\left[m_2,m_{\tilde{l}}^{i1}\right]+\\
& \frac{1}{192}\left(-2g_2^4\delta_{i1i4}\delta_{i2i3}+g_2^2\left(3g_1^2+g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{2,1,0}\left[m_2,m_{\tilde{l}}^{i2}\right]+\\
& \frac{1}{192}\left(-2g_2^4\delta_{i1i4}\delta_{i2i3}+g_2^2\left(3g_1^2+g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{2,2,-1}\left[m_2,m_{\tilde{l}}^{i2}\right]+\\
& \frac{1}{96}\left(-10g_2^4\delta_{i1i4}\delta_{i2i3}+g_2^2\left(-3g_1^2+5g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{3,1,-1}\left[m_2,m_{\tilde{l}}^{i2}\right]+\\
& \frac{1}{64}\left(2g_2^4\delta_{i1i4}\delta_{i2i3}+g_2^2\left(g_1^2-g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{4,1,-2}\left[m_2,m_{\tilde{l}}^{i2}\right]+\\
& \frac{1}{192}\left(-2g_2^4\delta_{i1i4}\delta_{i2i3}+g_2^2\left(3g_1^2+g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{2,1,0}\left[m_2,m_{\tilde{l}}^{i3}\right]+\\
& \frac{1}{192}\left(-2g_2^4\delta_{i1i4}\delta_{i2i3}+g_2^2\left(3g_1^2+g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{2,2,-1}\left[m_2,m_{\tilde{l}}^{i3}\right]+\\
& \frac{1}{96}\left(-10g_2^4\delta_{i1i4}\delta_{i2i3}+g_2^2\left(-3g_1^2+5g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{3,1,-1}\left[m_2,m_{\tilde{l}}^{i3}\right]+\\
& \frac{1}{64}\left(2g_2^4\delta_{i1i4}\delta_{i2i3}+g_2^2\left(g_1^2-g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{4,1,-2}\left[m_2,m_{\tilde{l}}^{i3}\right]+\\
& \frac{1}{192}\left(-2g_2^4\delta_{i1i4}\delta_{i2i3}+g_2^2\left(3g_1^2+g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{2,1,0}\left[m_2,m_{\tilde{l}}^{i4}\right]+\\
& \frac{1}{192}\left(-2g_2^4\delta_{i1i4}\delta_{i2i3}+g_2^2\left(3g_1^2+g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{2,2,-1}\left[m_2,m_{\tilde{l}}^{i4}\right]+\\
& \frac{1}{96}\left(-10g_2^4\delta_{i1i4}\delta_{i2i3}+g_2^2\left(-3g_1^2+5g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{3,1,-1}\left[m_2,m_{\tilde{l}}^{i4}\right]+\\
& \frac{1}{64}\left(2g_2^4\delta_{i1i4}\delta_{i2i3}+g_2^2\left(g_1^2-g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{4,1,-2}\left[m_2,m_{\tilde{l}}^{i4}\right]-\\
& \frac{1}{12}g_1^2\left(\overline{y_e}^{i4p}y_e^{i3p}\delta_{i1i2}+\overline{y_e}^{i2p}y_e^{i1p}\delta_{i3i4}\right)\text{LF}_{2,1,0}\left[m_{\tilde{e}}^p,\tilde{\mu}\right]+\\
& \frac{1}{24}g_1^2\left(\overline{y_e}^{i4p}y_e^{i3p}\delta_{i1i2}+\overline{y_e}^{i2p}y_e^{i1p}\delta_{i3i4}\right)\text{LF}_{2,2,-1}\left[m_{\tilde{e}}^p,\tilde{\mu}\right]+\\
& \frac{1}{24}g_1^2\left(\overline{y_e}^{i4p}y_e^{i3p}\delta_{i1i2}+\overline{y_e}^{i2p}y_e^{i1p}\delta_{i3i4}\right)\text{LF}_{3,1,-1}\left[m_{\tilde{e}}^p,\tilde{\mu}\right]+\\
& \frac{1}{96}g_1^2\left(-2g_2^2\delta_{i1i4}\delta_{i2i3}+\left(-g_1^2+g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{2,1,0}\left[m_{\tilde{l}}^{i1},m_1\right]+\\
& \frac{1}{192}g_1^2\left(2g_2^2\delta_{i1i4}\delta_{i2i3}+\left(g_1^2-g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{3,1,-1}\left[m_{\tilde{l}}^{i1},m_1\right]+\\
& \frac{1}{96}\left(2g_2^4\delta_{i1i4}\delta_{i2i3}-g_2^2\left(3g_1^2+g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{2,1,0}\left[m_{\tilde{l}}^{i1},m_2\right]+\\
& \frac{1}{192}\left(-2g_2^4\delta_{i1i4}\delta_{i2i3}+g_2^2\left(3g_1^2+g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{3,1,-1}\left[m_{\tilde{l}}^{i1},m_2\right]+\\
& \frac{1}{96}g_1^2\left(-2g_2^2\delta_{i1i4}\delta_{i2i3}+\left(-g_1^2+g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{2,1,0}\left[m_{\tilde{l}}^{i2},m_1\right]+\\
& \frac{1}{192}g_1^2\left(2g_2^2\delta_{i1i4}\delta_{i2i3}+\left(g_1^2-g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{3,1,-1}\left[m_{\tilde{l}}^{i2},m_1\right]+\\
& \frac{1}{96}\left(2g_2^4\delta_{i1i4}\delta_{i2i3}-g_2^2\left(3g_1^2+g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{2,1,0}\left[m_{\tilde{l}}^{i2},m_2\right]+\\
& \frac{1}{192}\left(-2g_2^4\delta_{i1i4}\delta_{i2i3}+g_2^2\left(3g_1^2+g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{3,1,-1}\left[m_{\tilde{l}}^{i2},m_2\right]+\\
& \frac{1}{96}g_1^2\left(-2g_2^2\delta_{i1i4}\delta_{i2i3}+\left(-g_1^2+g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{2,1,0}\left[m_{\tilde{l}}^{i3},m_1\right]+\\
& \frac{1}{192}g_1^2\left(2g_2^2\delta_{i1i4}\delta_{i2i3}+\left(g_1^2-g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{3,1,-1}\left[m_{\tilde{l}}^{i3},m_1\right]+\\
& \frac{1}{96}\left(2g_2^4\delta_{i1i4}\delta_{i2i3}-g_2^2\left(3g_1^2+g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{2,1,0}\left[m_{\tilde{l}}^{i3},m_2\right]+\\
& \frac{1}{192}\left(-2g_2^4\delta_{i1i4}\delta_{i2i3}+g_2^2\left(3g_1^2+g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{3,1,-1}\left[m_{\tilde{l}}^{i3},m_2\right]+\\
& \frac{1}{96}g_1^2\left(-2g_2^2\delta_{i1i4}\delta_{i2i3}+\left(-g_1^2+g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{2,1,0}\left[m_{\tilde{l}}^{i4},m_1\right]+\\
& \frac{1}{192}g_1^2\left(2g_2^2\delta_{i1i4}\delta_{i2i3}+\left(g_1^2-g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{3,1,-1}\left[m_{\tilde{l}}^{i4},m_1\right]+\\
& \frac{1}{96}\left(2g_2^4\delta_{i1i4}\delta_{i2i3}-g_2^2\left(3g_1^2+g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{2,1,0}\left[m_{\tilde{l}}^{i4},m_2\right]+\\
& \frac{1}{192}\left(-2g_2^4\delta_{i1i4}\delta_{i2i3}+g_2^2\left(3g_1^2+g_2^2\right)\delta_{i1i2}\delta_{i3i4}\right)\text{LF}_{3,1,-1}\left[m_{\tilde{l}}^{i4},m_2\right]+\\
& \frac{1}{24}g_1^2\left(\overline{y_e}^{i4p}y_e^{i3p}\delta_{i1i2}+\overline{y_e}^{i2p}y_e^{i1p}\delta_{i3i4}\right)\text{LF}_{2,1,0}\left[\tilde{\mu},m_{\tilde{e}}^p\right]+\\
& \frac{1}{48}\left(\overline{y_e}^{i4p}\left(-y_e^{i3p}\left(g_1^2-3g_2^2\right)\delta_{i1i2}-6g_2^2y_e^{i1p}\delta_{i2i3}\right)+\right.\\
& \left.\overline{y_e}^{i2p}\left(-6g_2^2y_e^{i3p}\delta_{i1i4}-y_e^{i1p}\left(g_1^2-3g_2^2\right)\delta_{i3i4}\right)\right)\text{LF}_{3,1,-1}\left[\tilde{\mu},m_{\tilde{e}}^p\right]+\\
& \frac{1}{48}\left(\overline{y_e}^{i4p}\left(y_e^{i3p}\left(g_1^2-g_2^2\right)\delta_{i1i2}+2g_2^2y_e^{i1p}\delta_{i2i3}\right)+\right.\\
& \left.\overline{y_e}^{i2p}\left(2g_2^2y_e^{i3p}\delta_{i1i4}+y_e^{i1p}\left(g_1^2-g_2^2\right)\delta_{i3i4}\right)\right)\text{LF}_{4,1,-2}\left[\tilde{\mu},m_{\tilde{e}}^p\right]+\\
& \frac{1}{64}g_1^4\,\text{LF}_{2,1,1,-1}\left[m_1,m_{\tilde{l}}^{i1},m_{\tilde{l}}^{i3}\right]\delta_{i1i2}\delta_{i3i4}+\frac{1}{32}g_1^4m_1^2\,\text{LF}_{2,1,1,0}\left[m_1,m_{\tilde{l}}^{i1},m_{\tilde{l}}^{i3}\right]\delta_{i1i2}\delta_{i3i4}+\\
& \frac{1}{64}g_1^4\,\text{LF}_{2,1,1,-1}\left[m_1,m_{\tilde{l}}^{i2},m_{\tilde{l}}^{i4}\right]\delta_{i1i2}\delta_{i3i4}+\frac{1}{32}g_1^4m_1^2\,\text{LF}_{2,1,1,0}\left[m_1,m_{\tilde{l}}^{i2},m_{\tilde{l}}^{i4}\right]\delta_{i1i2}\delta_{i3i4}+\\
& \frac{1}{16}g_2^4\,\text{LF}_{2,1,1,-1}\left[m_2,m_{\tilde{l}}^{i1},m_{\tilde{l}}^{i2}\right]\delta_{i1i4}\delta_{i2i3}-\frac{1}{8}g_2^4m_2^2\,\text{LF}_{2,1,1,0}\left[m_2,m_{\tilde{l}}^{i1},m_{\tilde{l}}^{i2}\right]\delta_{i1i4}\delta_{i2i3}+\\
& \frac{1}{64}g_2^4\,\text{LF}_{2,1,1,-1}\left[m_2,m_{\tilde{l}}^{i1},m_{\tilde{l}}^{i3}\right]\delta_{i1i2}\delta_{i3i4}+\frac{5}{32}g_2^4m_2^2\,\text{LF}_{2,1,1,0}\left[m_2,m_{\tilde{l}}^{i1},m_{\tilde{l}}^{i3}\right]\delta_{i1i2}\delta_{i3i4}+\\
& \frac{1}{64}g_2^4\,\text{LF}_{2,1,1,-1}\left[m_2,m_{\tilde{l}}^{i2},m_{\tilde{l}}^{i4}\right]\delta_{i1i2}\delta_{i3i4}+\frac{5}{32}g_2^4m_2^2\,\text{LF}_{2,1,1,0}\left[m_2,m_{\tilde{l}}^{i2},m_{\tilde{l}}^{i4}\right]\delta_{i1i2}\delta_{i3i4}+\\
& \frac{1}{16}g_2^4\,\text{LF}_{2,1,1,-1}\left[m_2,m_{\tilde{l}}^{i3},m_{\tilde{l}}^{i4}\right]\delta_{i1i4}\delta_{i2i3}-\frac{1}{8}g_2^4m_2^2\,\text{LF}_{2,1,1,0}\left[m_2,m_{\tilde{l}}^{i3},m_{\tilde{l}}^{i4}\right]\delta_{i1i4}\delta_{i2i3}+\\
& \frac{1}{8}\overline{y_e}^{i2p}\overline{y_e}^{i4r}y_e^{i1r}y_e^{i3p}\,\text{LF}_{2,1,1,-1}\left[\tilde{\mu},m_{\tilde{e}}^p,m_{\tilde{e}}^r\right]+\\
& \frac{1}{16}g_1^2g_2^2\,\text{LF}_{1,1,1,1,-1}\left[m_1,m_2,m_{\tilde{l}}^{i1},m_{\tilde{l}}^{i2}\right]\delta_{i1i4}\delta_{i2i3}+\\
& \frac{1}{8}m_1m_2g_1^2g_2^2\,\text{LF}_{1,1,1,1,0}\left[m_1,m_2,m_{\tilde{l}}^{i1},m_{\tilde{l}}^{i2}\right]\delta_{i1i4}\delta_{i2i3}-\\
& \frac{1}{32}g_1^2g_2^2\,\text{LF}_{1,1,1,1,-1}\left[m_1,m_2,m_{\tilde{l}}^{i1},m_{\tilde{l}}^{i3}\right]\delta_{i1i2}\delta_{i3i4}-\\
& \frac{1}{16}m_1m_2g_1^2g_2^2\,\text{LF}_{1,1,1,1,0}\left[m_1,m_2,m_{\tilde{l}}^{i1},m_{\tilde{l}}^{i3}\right]\delta_{i1i2}\delta_{i3i4}-\\
& \frac{1}{32}g_1^2g_2^2\,\text{LF}_{1,1,1,1,-1}\left[m_1,m_2,m_{\tilde{l}}^{i2},m_{\tilde{l}}^{i4}\right]\delta_{i1i2}\delta_{i3i4}-\\
& \frac{1}{16}m_1m_2g_1^2g_2^2\,\text{LF}_{1,1,1,1,0}\left[m_1,m_2,m_{\tilde{l}}^{i2},m_{\tilde{l}}^{i4}\right]\delta_{i1i2}\delta_{i3i4}+\\
& \frac{1}{16}g_1^2g_2^2\,\text{LF}_{1,1,1,1,-1}\left[m_1,m_2,m_{\tilde{l}}^{i3},m_{\tilde{l}}^{i4}\right]\delta_{i1i4}\delta_{i2i3}+\\
& \frac{1}{8}m_1m_2g_1^2g_2^2\,\text{LF}_{1,1,1,1,0}\left[m_1,m_2,m_{\tilde{l}}^{i3},m_{\tilde{l}}^{i4}\right]\delta_{i1i4}\delta_{i2i3}\Bigg)
\end{aligned}
$$